# Heet Mehta

MACHINE LEARNING DEVELOPER & FULL-STACK ENGINEER

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## WORK EXPERIENCE

### CHARUSAT University (CSPIT)

MACHINE LEARNING INTERN

*Jan 2026 – Apr 2026 | Anand, Gujarat*

Python · TensorFlow · scikit-learn · NumPy · Pandas · Matplotlib · Jupyter

- Trained & evaluated supervised ML models (**Random Forest, SVM, Logistic Regression**) on structured research datasets; improved **F1 score by 12%** through systematic hyperparameter tuning and 5-fold cross-validation.
- Engineered end-to-end **feature engineering** pipelines (null imputation, scaling, one-hot encoding), **reducing data prep time ~40%**; produced ROC-AUC, precision-recall and confusion matrix reports for faculty stakeholders.

## PROJECTS

### TrustFin-SLM

TRUSTWORTHY LLM · FINANCE GitHub ↗

Jul 2026 – Present

Python · Hugging Face · Qwen3-1.7B · QLoRA · FinQA · FiQA · FinTrust · Evaluation Metrics

- Fine-tuned **Qwen3-1.7B** with balanced joint **QLoRA** across 16 baseline experiments, selecting the backbone on validation data only (**FinQA execution score 0.675**, **FiQA $R^2$ 0.730**).
- Built a reproducible 4-benchmark evaluation pipeline (FinQA, FiQA, FOMC, FinTrust) with **SHA-256 frozen data contracts** and **68 validation gates**, keeping protected test sets out of training and threshold tuning.
- Proposed **EVG-CAS** abstention method: a frozen threshold cut protected-test answered-error by **31.0% at 80% coverage**, validated with paired McNemar tests and bootstrap CIs; research paper draft in progress.

### Code Pie

COMPETITIVE PROGRAMMING PLATFORM GitHub ↗

Mar 2026 – Present

Node.js · React · TypeScript · PostgreSQL · Redis · Bull · Docker · Nginx · PgBouncer

- Part of an **18-contributor** team (formerly BinaryBattles); owned the **contest module and the architecture behind it**, including submission-queue capacity and rate thresholds across Redis + Bull queues and isolated Docker workers.
- Scaled to a distributed 3-node setup (Nginx load balancer, PgBouncer, Redis workers) sustaining **1,000+ requests per second** across the contest workflow: submission queue stages (worker runner, Redis queue/cache, PostgreSQL) and contest-page APIs (problem statements, results, submission history).
- Built the contest workspace on **Monaco Editor** with C/C++, Python and SQL judging; in-progress code persists in **localStorage** across reloads.

### AutoDevs

OPEN SOURCE CLI · PRODUCT HUNT Live ↗ GitHub ↗

2025 – Present

Go · Node.js · TypeScript · GitHub API · npm · Homebrew · Scoop · Docker

- Built a **cross-platform developer-environment bootstrapper** in Go that detects a project's stack and installs missing runtimes, dependencies and tools in one command: 40+ packages, 9 dev-role profiles, 3 OS targets.
- Published on npm, Homebrew, Scoop & Docker and **featured on Product Hunt**; its GitHub Scanner scores language/framework confidence from public repos using frequency-weighted heuristics and outputs a skill matrix with upgrade paths.

### Coastal Guardian

DEEP LEARNING · COMPUTER VISION GitHub ↗

2025

Python · TensorFlow · CNN · Neural Networks · NASA Satellite Imagery · scikit-learn

- Designed a **CNN deep learning neural network** on NASA satellite imagery to classify 4 coastal threat categories (erosion, pollution, algal bloom, clear) using **ResNet50 transfer learning** and augmentation on imbalanced classes.
- Achieved **87%+ validation accuracy** on 2,400 image patches via systematic statistical model evaluation (confusion matrix, ROC-AUC, precision-recall); visualized results with Matplotlib/Seaborn for research stakeholders.

## OPEN SOURCE · CERTIFICATIONS · LEADERSHIP

- **AirLLM** | PR #339 GitHub ↗ — Fixed layer-count parsing for nested checkpoint prefixes (VLM/multimodal models), removing a `ValueError` crash when splitting checkpoints; added unit tests across supported architectures.
- **Ollama** | Mar 2026 — PR #14992 GitHub ↗ — Removed duplicate TopLogprobs validation dead code across GenerateHandler and ChatHandler in a production Go LLM inference engine.
- **Open Source Club, CHARUSAT** Website ↗ — Developer of the club website and ML tutor for club members.
- **Analytics with Python** – Coursera (2026) · **Algorithm Design** – Coursera (2026) · **Foundations of DSA** – Packt (2024) · **Coordinator**, C Titans Hackathon (50+ participants) & SUPERNOVA — CHARUSAT.

## SKILLS

**ML / AI**

TensorFlow · scikit-learn · Hugging Face · QLoRA · LLM Fine-tuning · Deep Learning · NumPy · Pandas · Feature Engineering · Evaluation Metrics (F1, ROC-AUC, McNemar) · LLM APIs · Prompt Engineering

**LANGUAGES**

Python · Go · TypeScript · JavaScript · C · C++ · Java

**FRONTEND & BACKEND**

React · Next.js · Node.js · Express.js · REST APIs · WebSocket

**DATABASES**

PostgreSQL · MongoDB · Redis · PgBouncer

**DEVOPS**

Docker · Nginx · GitHub Actions · CI/CD · Linux

**MATHEMATICS**

Probability & Statistics · Linear Algebra · Calculus · Discrete Math · Statistical Modeling

## EDUCATION

### CHARUSAT UNIVERSITY

*B.Tech – CSE (CSPIT)*

Anand, Gujarat · 2024 – 2028

**CGPA: 7.86 / 10**

Probability & Statistics · DSA · DBMS · Networks · Python for DS

**HSC - SCIENCE** GSEB 2024
**95.40 Percentile**

**SSC** GSEB 2022
**98.45 Percentile**

## PROFESSIONAL SUMMARY

*Third-year B.Tech CSE student at CHARUSAT (CGPA 7.86) building and evaluating LLM systems. Fine-tuned small language models with QLoRA, designed abstention-based trust evaluation, and shipped distributed full-stack systems. Open-source contributor to Ollama and AirLLM. Seeking an ML Developer internship.*

## EXTRA CURRICULARS

- Contributor — **Ollama**, **AirLLM** open-source LLM projects.
- Developer & ML tutor — **Open Source Club** CHARUSAT.
- Featured on **Product Hunt** — AutoDevs CLI.
- Hackathon coordinator — **50+ participants**.
- Active competitive programmer on **LeetCode**.

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